https://www.iitg.ac.in/cseweb/vlab/Digital-System-Lab/fa_mux.php?id=10

Full Adder

Truth Table				
Inputs			Outputs	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full Adder using 4 to 1 Multiplexer:

S(A,B,Cin)= ∑(1,2,4,7)

Cout(A,B,Cin)= ∑(3,5,6,7)

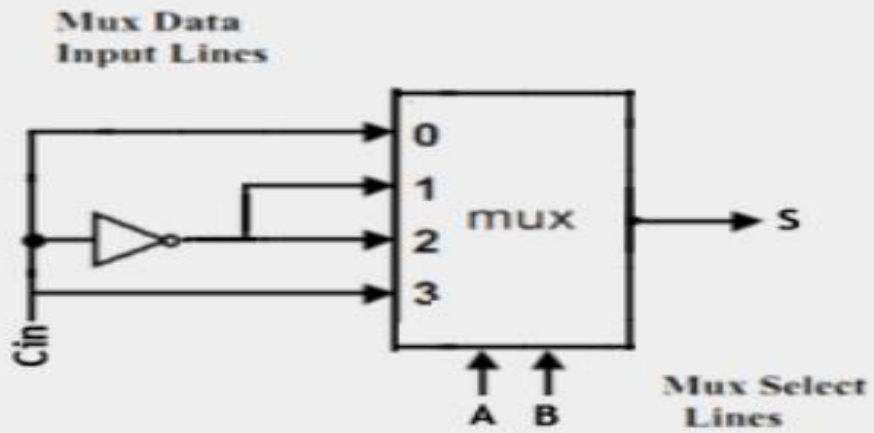
For sum out:

Cin \ AB	00	01	10	11
0	0	1	1	0
1	1	0	0	1
	Cin	C'in	C'in	Cin

For carry out:

Cin \ AB	00	01	10	11
0	0	0	0	1
1	0	1	1	1
	0	Cin	Cin	1

Logic digram for Sum out



Logic digram for Carry out

